

$$1) \quad x_2[m] = x_1[(m-4) \bmod 8] \Rightarrow x_2[m] = x_1[m+4]$$

$$x_2[m] = x_1[m-m_0] \Rightarrow x_2[k] = e^{-j\frac{2\pi}{N} k m_0} x_1[k]; \quad m_0 = 4; N = 8$$

$$x_2[k] = e^{-j\frac{2\pi}{8} k \cdot 4} x_1[k] = e^{-j\pi k} x_1[k]$$

$$2) \quad a) \quad x[0] = 4; x[1] = 1; x[2] = 1; x[3] = 2$$

$$X[k] = \sum_{m=0}^3 x[m] e^{-j\frac{2\pi}{4} k m} \Rightarrow$$

$$\Rightarrow X[0] = 4 + 1 + 1 + 2 = 8; \quad X[1] = 4 + e^{-j\frac{\pi}{4} \cdot 1} + 2e^{-j\frac{\pi}{4} \cdot 2} + 2e^{-j\frac{2\pi}{4} \cdot 3} =$$

$$= 4 + e^{-j\frac{\pi}{2}} + 2e^{-j\pi} + 2e^{-j\frac{3\pi}{2}};$$

$$X[2] = 4 + e^{-j\frac{\pi}{2} \cdot 2} + 2e^{-j\pi \cdot 2} + 2e^{-j\frac{3\pi}{2} \cdot 3} = 4 + e^{-j\pi} + 2e^{-j2\pi} + 2e^{-j\frac{9\pi}{2}}$$

$$X[3] = 4 + e^{-j\frac{3\pi}{4} \cdot 3} + 2e^{-j\frac{3\pi}{4} \cdot 3 \cdot 2} + 2e^{-j\frac{3\pi}{4} \cdot 3 \cdot 3} = 4 + e^{-j\frac{9\pi}{4}} + 2e^{-j\frac{9\pi}{2}} + 2e^{-j\frac{27\pi}{4}}$$

$$b) \quad y[m] = \sum_{n=0}^3 x[n] x[(m-n) \bmod 4], \quad m = 0, 1, 2, 3$$

$$y[0] = x[0]x[0] + x[1]x[3] + x[2]x[2] + x[3]x[1] = 4 \cdot 4 + 1 \cdot 2 + 2 \cdot 2 + 2 \cdot 1 = 16 + 4 + 4 + 2 = 24$$

$$y[1] = x[0]x[1] + x[1]x[0] + x[2]x[3] + x[3]x[2] = 4 \cdot 1 + 1 \cdot 4 + 2 \cdot 2 + 2 \cdot 2 = 16$$

$$y[2] = x[0]x[2] + x[1]x[1] + x[2]x[0] + x[3]x[3] = 4 \cdot 2 + 1 \cdot 1 + 2 \cdot 4 + 2 \cdot 2 = 21$$

$$y[3] = x[0]x[3] + x[1]x[2] + x[2]x[1] + x[3]x[0] = 4 \cdot 2 + 1 \cdot 2 + 2 \cdot 1 + 2 \cdot 4 = 20$$

$$y = [24, 16, 21, 20]$$

$$c) X[k] = X[k] \cdot X[k] = X[k]^2 \Rightarrow X[0] = 9^2 = 81, X[1] = (2+j)^2 = 4 + 4j - 1 = 3 + 4j$$

$$X[2] = 3^2 = 9, X[3] = (2-j)^2 = 4 - 4j - 1 = 3 - 4j \Rightarrow X = [81, 3+4j, 9, 3-4j]$$

$$y[m] = \frac{1}{4} \sum_{k=0}^3 X[k] e^{j \frac{\pi}{2} km}; \quad m=0,1,2,3; e^{j \frac{\pi}{2}} = 1; e^{j \pi} = -1; e^{j \frac{3\pi}{2}} = -j$$

$$y[0] = \frac{1}{4} (81 + (3+4j) + 9 + (3-4j)) = \frac{1}{4} (81 + 3 + 9 + 3) = \frac{96}{4} = 24$$

$$y[1] = \frac{1}{4} (81 + (3+4j)j + 9(-1) + (3-4j)(-j)) = \frac{64}{4} = 16$$

$$y[2] = \frac{1}{4} (81 - (3+4j) + 9 - (3-4j)) = \frac{1}{4} (81 - 3 - 4j + 9 - 3 + 4j) = \frac{84}{4} = 21$$

$$y[3] = \frac{1}{4} \cdot (81 + (3+4j)(-j) + 9(-1) + (3-4j)j) = \frac{80}{4} = 20$$

$$y[m] = [24, 16, 21, 20]$$

$$d) L_{\text{lim}} = 4 + 4 - 1 = 7 \Rightarrow N_{\text{min}} = 7$$

$$m=0 \Rightarrow y_{\text{lim}}[0] = 4 \cdot 4 = 16; m=1 \Rightarrow y_{\text{lim}}[1] = 4 \cdot 1 + 1 \cdot 4 = 8; m=2 \Rightarrow y_{\text{lim}}[2] = 4 \cdot 2 + 1 \cdot 1 + 1 \cdot 4 = 17$$

$$m=3 \Rightarrow y_{\text{lim}}[3] = 4 \cdot 2 + 1 \cdot 2 + 1 \cdot 1 + 1 \cdot 4 = 20; m=4 \Rightarrow y_{\text{lim}}[4] = 1 \cdot 2 + 1 \cdot 2 + 1 \cdot 1 = 8; m=5 \Rightarrow y_{\text{lim}}[5] = 2^2 + 2^2 = 8$$

$$m=6 \Rightarrow y_{\text{lim}}[6] = 2 \cdot 2 = 4$$

$$e) 3) X_3[m] = \sum_{k=0}^7 x_1[k] \cdot x_2[(m-k) \bmod 8] \Rightarrow X_3[2] = \sum_{k=0}^7 x_1[k] \cdot x_2[(2-k) \bmod 8] \quad k[0,1,\dots,7]$$

$$X_3[2] = 1 \cdot 2 + 1 \cdot 3 + 0 + 0 + 0 + 0 + 0 + 6 = 5 \Rightarrow X[2] = 5$$

Processamento de sinais digitais $\Rightarrow$ Victor E. de L. Guerra, Trabalho DFT - p. 2

$$4) y_1[n] = \sum_{m=-\infty}^{\infty} x[m]u[m]; \quad n \in \{0, 1, 2, 3\}$$

$$\Rightarrow x[0]=1, x[1]=0, x[2]=2, x[3]=2, x[4]=0, x[5]=1$$

$$\Rightarrow x_1[0]=4, x_1[1]=1, x_1[2]=2, x_1[3]=2$$

$$\Rightarrow x_1[0] = x[0] + x[4] + x[8] + \dots = 1 + 0 + 0 + \dots = 1 + 0 \Rightarrow x_1[0] = 4 \Rightarrow 4 = 1 + 0 \Rightarrow 0 = 3$$

$$5) a) x_1[0]=1, x_1[1]=-2, x_1[2]=-1, x_1[3]=3$$

$$x_2[1]=2, x_2[4]=-1, x_2[5]=1$$

$$y[n] = \sum_{k=0}^4 x_1[k] x_2[(n-k) \bmod 5] \Rightarrow x_1^{(n)} = [1, 2, 0, 0, -1]$$

$$n=0: k=0 \Rightarrow 1 \cdot x_2[0] = 1 \cdot 1 = 1; k=1 \Rightarrow -2 \cdot x_2[4] = -2 \cdot (-1) = 2 \Rightarrow y[0] = 1 + 2 = 3$$

$$k=2 \Rightarrow -1 \cdot x_2[3] = 0; k=3 \Rightarrow -1 \cdot x_2[2] = 0; k=4 \Rightarrow 0 \cdot x_2[1] = 0$$

$$n=1: k=0 \Rightarrow 1 \cdot x_2[1] = 1 \cdot 2 = 2; k=1 \Rightarrow -2 \cdot x_2[0] = -2 \cdot 1 = -2; k=2 \Rightarrow -1 \cdot x_2[4] = -1 \cdot (-1) = 1$$

$$k=3 \Rightarrow 3 \cdot x_2[3] = 0; k=4 \Rightarrow 0 \cdot x_2[2] = 0 \Rightarrow y[1] = 2 - 2 + 1 = 1$$

$$n=2: k=0 \Rightarrow 1 \cdot x_2[2] = 0; k=1 \Rightarrow -2 \cdot x_2[1] = -4; k=2 \Rightarrow -1 \cdot x_2[0] = -1; k=3 \Rightarrow 3 \cdot x_2[4] = -3$$

$$k=4 \Rightarrow 0 \cdot x_2[3] = 0 \Rightarrow y[2] = -4 - 1 - 3 = -8$$

$$n=3: k=0 \Rightarrow 1 \cdot x_2[3] = 0; k=1 \Rightarrow -2 \cdot x_2[2] = 0; k=2 \Rightarrow -1 \cdot x_2[1] = -2; k=3 \Rightarrow 3 \cdot x_2[0] = 3 \cdot 1 = 3$$

$$k=4 \Rightarrow 0 \cdot x_2[4] = 0 \Rightarrow y[3] = -2 + 3 = 1$$

$$n=4: k=0: 1 \cdot x_2[4] = -1; k=1 \Rightarrow -2 \cdot x_2[3] = 0; k=2 \Rightarrow -1 \cdot x_2[2] = 0; k=3 \Rightarrow 3 \cdot x_2[1] = 6; k=4 \Rightarrow 0 \cdot x_2[0] = 0$$

$$y[4] = -1 + 6 = 5$$

$$y[n] = [3, 1, -8, 1, 5]$$

$$b) N \geq L_1 + L_2 - 1 \Rightarrow L_1 = 4, L_2 = 6 \Rightarrow L = 4 + 6 - 1 = 9$$

$$N_{min} = 9 \Rightarrow y[n] = \sum_{k=0}^n x_1[k] \cdot x_2[n-k]$$

$$n=0: y[0] = x_1[0] \cdot x_2[0] = 1 \cdot 0 = 0$$

$$n=1: y[1] = x_1[0] x_2[1] + x_1[1] x_2[0] = 1 \cdot 2 + (-1) \cdot 0 = 2$$

$$n=2: y[2] = x_1[0] x_2[2] + x_1[1] x_2[1] + x_1[2] x_2[0] = -2 \cdot 2 = -4$$

$$n=3: y[3] = x_1[0] x_2[3] + x_1[1] x_2[2] + x_1[2] x_2[1] + x_1[3] x_2[0] = -1 \cdot 2 = -2$$

$$n=4: y[4] = x_1[0] x_2[4] + x_1[1] x_2[3] + x_1[2] x_2[2] + x_1[3] x_2[1] + x_1[4] x_2[0] = 1 \cdot (-2) + (3 \cdot 2) = 6 - 2 = 4$$

$$n=5: y[5] = x_1[1] x_2[4] + x_1[2] x_2[3] + x_1[3] x_2[2] = -2 + 1 = -1$$

$$n=6: y[6] = x_1[2] x_2[4] + x_1[3] x_2[3] = -1 \cdot 3 = -3$$

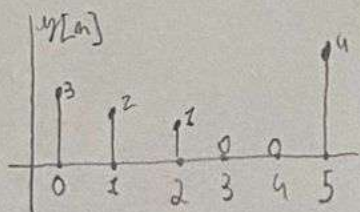
$$n=7: y[7] = x_1[3] x_2[4] = 3 \cdot 1 = 3$$

$$y_{lim} = [0, 2, -4, -2, 4, -1, -3, 3]$$

$$6) a) x[k] = W_N^{m_0 k} X[k] = e^{-j 2 \pi \frac{m_0}{N} \cdot k m_0} X[k] \Rightarrow y[m] = x[(m - m_0) \bmod N]$$

$$N=6, m_0=5 \Rightarrow y[m] = x[(m-5) \bmod 6] \Rightarrow x[m] = [4, 3, 2, 1, 0, 0]$$

$$y[0] = x[1] = 3; y[1] = x[2] = 2; y[2] = x[3] = 1; y[3] = x[4] = 0; y[4] = x[5] = 0; y[5] = x[0] = 4$$



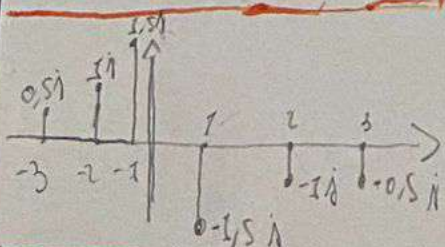
$$b) W[m] = \frac{x[m] - x^*[m]}{2j}$$

$$x^*[m] = x[-m]$$

$$W[m] = x[m] - x[-m] = \frac{x[m] - x[-m]}{2j} = -j \frac{x[m] - x[-m]}{2}$$

$$c) X[k] = 4 + 3e^{-j \frac{\pi}{6} k} + 2e^{-j \frac{\pi}{6} 2k} + e^{-j \frac{\pi}{6} 3k}$$

$$Q[k] = X[2k+1] \Rightarrow Q[k] = 4 + 3(e^{-j \frac{\pi}{3}})^k \cdot e^{-j \frac{\pi}{3}} + 2(e^{-j \frac{\pi}{3}})^{2k} \cdot e^{-j \frac{2\pi}{3}} + e^{-j \pi} \Rightarrow Q[k] = 3\delta[k] + 3(e^{-j \frac{\pi}{3}})^k \delta[k-1]$$



Questão 2(d)

Código MATLAB para calcular a DFT de um sinal $x[n]$, realizar a convolução circular de 4 pontos (direta e via DFT), e comparar com a convolução linear.

```
1 x = [4 1 2 2];           % x[0], x[1], x[2], x[3]
2
3 % (a) DFT de 4 pontos
4 X = fft(x,4);
5
6 % exibir X
7 disp('X = '); disp(X);
8
9 % (b) convolucao circular de 4 pontos (direta)
10 y_circ = zeros(1,4);
11 for n=0:3
12     s = 0;
13     for m=0:3
14         s = s + x(m+1)*x(mod(n-m,4)+1);
15     end
16     y_circ(n+1) = s;
17 end
18 disp('y circular (direta) = '); disp(y_circ);
19
20 % (c) via DFT
21 Y = X .* X;
22 y_ifft = ifft(Y,4);
23 disp('y via DFT/IDFT = '); disp(round(y_ifft)); % round para remover erros numéricos
24
25 % (d) convolucao linear e N minimo
26 y_lin = conv(x,x); % length 7
27 disp('y linear = '); disp(y_lin);
28 % N minimo = length(y_lin) = 7
```

Questão 5(c)

Código MATLAB para calcular a convolução circular de 5 pontos entre $x_1[n]$ e $x_2[n]$, determinar o menor N que iguala a convolução circular à linear, e exibir os resultados.

```
1 % definir x1 e x2 segundo leitura do enunciado
2 % x1(n) não nulo em n=0..3
3 x1 = [1, -2, -1, 3];          % comprimento 4
4
5 % x2 tem valores em n=1,4,5 -> montamos vetor 0..5 (length 6)
6 x2_lin = [0, 2, 0, 0, -1, 1]; % indices 0..5
7
8 % (a) Convolucao circular N = 5
9 % construir vetores mod 5 (n=0..4)
10 x1_5 = [1, -2, -1, 3, 0];
11 x2_5 = [ 1, 2, 0, 0, -1 ]; % note: x2(5) foi somado em x2_5(0)
12
13 y_circ5 = cconv(x1_5, x2_5, 5) % cconv faz convolucao circular de comprimento N
14
15 % (b) Menor N para igualar a convolucao linear
16 Nmin = length(x1) + length(x2_lin) - 1 % deve ser 9
17 y_lin = conv(x1, x2_lin)                % convolucao linear (length 9)
18
19 % (opcional) mostrar ambos lado a lado
20 disp('circular N=5:'); disp(y_circ5);
21 disp('linear conv:'); disp(y_lin);
```